

Zeta Functions of Finite Graphs and Coverings

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Three different zeta functions are attached to a finite connected, possibly irregular graph X . They originate with a zeta function of Ihara which is an analogue of Riemann's as well as Selberg's zeta function. The three zeta functions are associated to one vertex variable, two variables for each edge, and $2r(2r-1)$ path variables, respectively. Here r is the number of generators of the fundamental group of X . We show how to specialize the variables of the last two zeta functions to obtain the first and we give elementary proofs of generalizations of Ihara's formula which says that the zeta function for a regular graph is the reciprocal of a polynomial. Many examples of covering graphs are also considered. © 1996 Academic Press, Inc.

1. INTRODUCTION

The aim of this paper is to give an elementary discussion of Ihara-type zeta functions and their generalizations for finite, connected, possibly irregular graphs. We will consider zeta functions with vertex variables, zeta functions with edge variables, and zeta functions with path variables, the latter appearing here for the first time.

Later in this section, we consider Ihara's zeta function $Z_X(u)$ for a finite connected $(q+1)$ -regular graph X . Ihara's theorem [7] says that $Z_X(u)^{-1}$ is an explicit polynomial in u . We deduce a number of corollaries, including the fact that if Y is a finite, connected $(q+1)$ -regular covering graph of X , then $Z_X(u)^{-1}$ divides $Z_Y(u)^{-1}$. Various covers of the tetrahedron K_4 are considered.

Section 2 is devoted to an elementary proof of the generalization of Ihara's theorem to finite possibly irregular graphs X . The discussion is simpler than that of Bass [1], Hashimoto [5], or Venkov and Nikitin [17]. It involves only a generalization of the argument sketched in Sunada [14] for regular graphs. Extra innovations are, however, needed for

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irregular graphs. Surprisingly these innovations lead to a simplification of the proof. Our discussion does not require non-commutative determinants as in [1], bipartite graphs as in [5], or Selberg's trace formula as in [17]. In fact, the trace formula seems only to have been worked out for regular graphs, but our results suggest that it must be possible to do a more general trace formula. Section 2 closes with some examples:

- (1) A Klein 4-group cover Y of a subgraph X of K_4 and the quadratic subcovers Q_1 , Q_2 , and Q_3 between Y and X ;
- (2) A six-fold (S_3) cover of K_4 and some of its subcovers.

Section 3 concerns multivariable zeta functions $\zeta_X(u_1, \dots, u_m)$, where m is twice the number of edges of X . If you specialize the variables $u_j = u$, for all $j = 1, \dots, m$, you get the Ihara zeta function of X . And if you specialize to 0 the pair of variables corresponding to an edge e of X , you get $\zeta_{X'}$, for the subgraph X' of X obtained by deleting edge e . Finally, if you specialize the two variables corresponding to each edge to be equal, you obtain Hashimoto's zeta function of $m/2$ variables considered in [5]. The first theorem of this section is an analogue of Ihara's theorem for ζ_X . Again our discussion is completely elementary and, amazingly enough, even simpler than that of Section 2. Once more, we do not need to look at bipartite graphs as in Hashimoto [5] or non-commutative determinants as in Bass [1]. However, in essence, the proof is a recasting of Bass's proof. The same methods will be used to show that our multivariable path zeta function is a rational function. The section ends with an example showing how the specialization results alluded to above work out for a subgraph of K_4 .

In Section 4 we consider a new zeta function with $(2r)^2 - 2r$ variables, where r is the rank of the fundamental group Γ of the graph; i.e., the number of generators of Γ . This function is the same for all rank r graphs and may thus be thought of as a universal rank r zeta function. Amazingly enough, there is again a way to specialize the variables to obtain the multivariable zeta function of Section 3 and, therefore, the Ihara zeta function.

In a later paper we will generalize our method to Artin L -functions of graphs as considered by Hashimoto [6].

In the rest of this section we survey what is known about zeta functions of finite mostly regular graphs.

Many generalizations of *Riemann's zeta function*

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} \quad \text{for } \operatorname{Re} s > 1. \quad (1.1)$$

have been found to be useful: Dirichlet L -functions, Dedekind zeta functions, Artin L -functions, Selberg zeta functions, Ruelle zeta functions.

References are Davenport [3], Lang [8], Ruelle [11], and the article on zeta functions in [13].

In what follows we want to study zeta functions associated to a finite connected graph X . Our basic reference for graph theory is Biggs [2]. Throughout the rest of this section, we will assume that the graph X is a $(q+1)$ -regular graph; i.e., there are $(q+1)$ edges emanating from each vertex. In later sections, X need not be regular.

The zeta functions associated to finite graphs by Ihara [7], Hashimoto [5], and others combine features of Riemann's zeta function, Artin L -functions, and Selberg's zeta function. They are defined by an Euler product and have analytic continuation to a meromorphic function with a functional equation. The Riemann hypothesis for zeta functions of regular graphs is sometimes true and sometimes false (as examples show). In this section we give a quick summary of the basic definitions. Other references are: Bass [1], Hashimoto [5, 6], Sunada [14, 15], and Venkov and Nikitin [17].

The Ihara zeta function is a product over primitive cycles in X without backtracking or tails. We need to define all these words. Here we follow Hashimoto [5] for the most part. A *cycle* C in X is a closed path. That is, C is a sequence of vertices $C = (v_1, v_2, \dots, v_m = v_1)$. Here $\nu(C) = \text{length of } C = m - 1 = \text{number of edges in } C$. We say that a cycle C has *backtracking* if $v_{i-1} = v_{i+1}$, for some i , $2 \leq i \leq m - 1$. Our product will be over *backtrack-less* or *proper* cycles C .

Every cycle C without backtracking determines a unique *tail-less* or *reduced*, backtrack-less cycle C^* obtained by removing the edges shown in Fig. 1. The cycle C^* will be *tail-less* iff $(C^*)^2$ has no backtracking.

There is an *equivalence* relation between backtrack-less, tail-less cycles $C = (v_1, \dots, v_m = v_1)$ and $D = (w_1, \dots, w_m = w_1)$. We say $C \sim D$ if $w_j = v_{j+k}$,

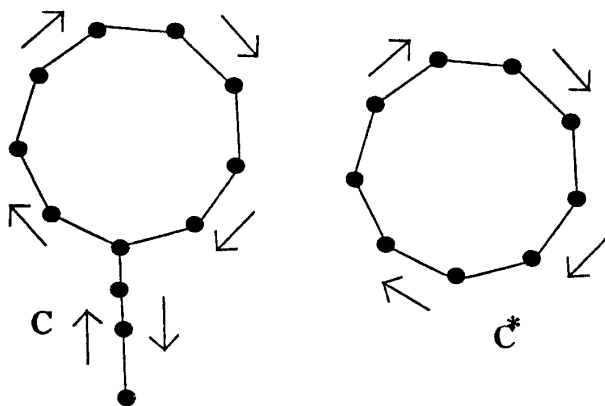


FIG. 1. Backtrack-less cycle C with tail and unique corresponding tail-less cycle C^* .

for all j . That is, we simply shift the origin k steps. Let $[C]$ denote the class of all backtrack-less, tail-less cycles D equivalent to C .

We define a backtrack-less, tail-less cycle C to be *primitive* if C is not obtained by going r times around some other cycle B ; i.e., $C \neq B^r$. Now we can define the Ihara zeta function.

DEFINITION. The *Ihara zeta function* associated to a finite connected graph X is defined to be a function of $u \in \mathbb{C}$ with u sufficiently small, by

$$Z_X(u) = \prod_{[C]} (1 - u^{v(C)})^{-1}. \quad (1.2)$$

The product is over equivalence classes of primitive closed backtrack-less, tail-less cycles $C = (v_1, \dots, v_m = v_1)$ of positive length m in X . Here $v(C) = m - 1 = \text{length of } C = \text{number of edges in } C$.

We should really be thinking as algebraic topologists would (see Massey [10, Chapter 6]). The *fundamental group* $\Gamma = \pi_1(X, p)$ of X , for a point $p \in X$, is the group of homotopy classes of cycles in X beginning and ending at p , under the product obtained from composition of paths (meaning first go around one path, next the other path). The Seifert-von Kampen theorem (Massey [10, p. 198]) says $\pi_1(X, p)$ is a free group of rank r , where r is the number of edges left out in order to obtain a spanning tree. A *tree* is a connected graph without cycles. A *spanning tree* in X is a tree which is a subgraph of X and which includes every vertex of X .

We have a 1-1 correspondence between conjugacy classes $\{C\}$ in $\Gamma = \pi_1(X, p)$ and equivalence classes of backtrack-less, tail-less cycles $[C]$ in X . If a closed path C starting and ending at p gives rise to a conjugacy class $\{C\}$ in Γ , we may take C in its homotopy class so that C has no backtracking. We then remove the tail from C so as to get a tail-less cycle C^* as above. The 1-1 correspondence referred to comes from the fact that the conjugacy class of C in Γ corresponds to the equivalence class of the backtrack-less, tail-less cycle C^* . With effort, it can be seen that change of C in its conjugacy class corresponds to a change of C^* in its equivalence class. In the other direction of the correspondence, given C^* , we grow a tail so as to reach p , thereby getting a path C giving rise to an element of the fundamental group. A different tail simply conjugates C . Another way of thinking of the correspondence is that the elements of the equivalence class of C^* are precisely the closed cycles of minimal length which are freely homotopic to C . Freely homotopic means the base point is not fixed.

The fundamental group is a free group of rank r and thus the centralizer of $C \neq 1$ in $\pi_1(X, p)$ is cyclic. Under the 1-1 correspondence between classes

$[C]$ of backtrack-less, tail-less cycles in X and conjugacy classes $\{C\}$ in Γ , primitive cycles C correspond to conjugacy classes $\{C\}$ in Γ such that the centralizer of C in Γ is generated by C .

Algebraic topology tells us that there is a *universal covering tree* $\tilde{X}$ (regular iff X is regular) and there is an action of the fundamental group $\Gamma = \pi_1(X, p)$ on $\tilde{X}$ such that $\tilde{X}/\Gamma \cong X$. You can also view the tree (in the $(p+1)$ -regular case) as coming from p -adic matrix groups. See Serre [12] and Trimble [16].

Let V be the set of vertices of X and E the set of edges. Then the *Euler characteristic* of X is

$$\chi(X) = |V| - |E| = 1 - r, \quad \text{where } r = \text{rank } \pi_1(X, p). \quad (1.3)$$

If X is a $(q+1)$ -regular graph with n vertices, this means

$$-\chi(X) = r - 1 = \frac{n(q-1)}{2}. \quad (1.4)$$

The number $r - 1$ will appear prominently in Theorem 1.

One moral of the preceding considerations is that we can rewrite the zeta-function (1.2) in the language of algebraic topology as

$$Z_X(u) = \sum_{\substack{\{C\} \text{ primitive conjugacy} \\ \text{classes in } \Gamma = \pi_1(X, p) \\ C \neq \text{identity}}} (1 - u^{v(C)})^{-1}. \quad (1.5)$$

Here $v(C)$ means the length of an element C^* in the equivalence class of tail-less, backtrack-less paths corresponding to $\{C\}$. Alternatively, $v(C)$ is the minimal length of all cycles freely homotopic to C . Note that we distinguish between C and C^{-1} in the product. This version of the zeta function will figure prominently in the last section.

SIMPLEST EXAMPLE. Let X be a cycle graph with n vertices. Then

$$Z_X(u) = (1 - u^n)^{-2}.$$

We get the power 2 because there are two primitive cycles, one going around clockwise once, and the other going around counterclockwise once.

In order to compute the zeta function easily in other cases when there are infinitely many primitive cycles, one needs the following theorem.

THEOREM 1 (Ihara [7]). *If X is a connected $(q+1)$ -regular graph with adjacency matrix A and $r = \text{rank}$ of the fundamental group (see (1.4)), then*

the Ihara zeta function defined by (1.2) is the reciprocal of a polynomial. More precisely, we have

$$Z_X^{-1}(u) = (1 - u^2)^{r-1} \det(I - Au + qu^2I). \quad (1.6)$$

A generalization of this result to possibly non-regular graphs will be proved in the next section.

In analogy with the *Riemann hypothesis* which says that if $0 < \operatorname{Re} s < 1$ and $\zeta(s) = 0$, then $\operatorname{Re}(s) = \frac{1}{2}$, we can make the following definitions.

DEFINITION. Suppose that X is a finite connected $(q+1)$ -regular graph. Set $u = q^{-s}$. $Z_X(q^{-s})$ is said to *satisfy the Riemann hypothesis* iff for

$$\operatorname{Re} s \in (0, 1), \quad Z_X(q^{-s}) = 0 \quad \text{implies} \quad \operatorname{Re} s = \frac{1}{2}. \quad (1.7)$$

The next definition was made by Lubotzky, Phillips, and Sarnak [9].

DEFINITION. A $(q+1)$ -regular graph is a *Ramanujan graph* iff for every eigenvalue λ of the adjacency matrix A such that $|\lambda| \neq q+1$, we have

$$|\lambda| \leq 2\sqrt{q}. \quad (1.8)$$

Ramanujan graphs make good communications networks (see Friedman [4]).

COROLLARY 1. $Z_X(q^{-s})$ satisfies the Riemann hypothesis iff X is a Ramanujan graph.

Proof. Let $\operatorname{spec}(A)$ be the set eigenvalues of the adjacency matrix A of X . These eigenvalues are real since A is symmetric. Then

$$Z_X^{-1}(u) = (1 - u^2)^{r-1} \prod_{\lambda \in \operatorname{spec}(A)} (1 - \lambda u + qu^2).$$

And $1 - \lambda u + qu^2 = (1 - \alpha u)(1 - \beta u)$, where $\alpha\beta = q$ and $\alpha + \beta = \lambda$ and

$$\alpha, \beta = \frac{\lambda \pm \sqrt{\lambda^2 - 4q}}{2}.$$

Thus $|\lambda| \leq 2\sqrt{q}$ iff α, β are complex conjugates with absolute value $\sqrt{q}$. If we write $q^s = \alpha, \beta$, with $q = \sigma + it$, we see that $\sigma = \frac{1}{2}$. When $\lambda = \pm(q+1)$, we have $q^s = \pm 1$ or $\pm q$, i.e., $\operatorname{Re}(s) = 0$ or 1 . Note that, except for $\lambda = \pm(q+1)$, we have $|\lambda| < q+1$ and, thus, $1 < |\alpha|, |\beta| < q$. The corresponding $q^s = \alpha$ or β then has $0 < \operatorname{Re}(s) < 1$. ■

Next we see that, as with many other zeta functions, there is a functional equation relating $Z_X(u)$ to $Z_X(1/qu)$. You just need to multiply $Z_X(u)$ $(1-u^2)^{r-1}$ by a polynomial $p(u)$ such that $p(1/qu^2) = \pm (qu^2)^{-n} p(u)$.

COROLLARY 2 (Functional equations). *Under the hypotheses of Theorem 1, we have the following functional equations of the Ihara zeta function among others:*

$$(1) \quad A_X(u) = (1-u^2)^{n/2+r-1} (1-q^2u^2)^{n/2} Z_X(u) = (-1)^n A_X(1/qu).$$

$$(2) \quad \xi_X(u) = (1+u)^{r-1} (1-u)^{r-1+n} (1-qu)^n Z_X(u) = \xi_X(1/qu).$$

$$(3) \quad \Xi_X(u) = (1-u^2)^{r-1} (1+qu^2)^n Z_X(u) = \Xi_X(1/qu).$$

Proof. We will prove (1) and leave the rest to the reader. To see (1), write

$$\begin{aligned} A_X(u) &= (1-u^2)^{n/2} (1-q^2u^2)^{n/2} \det(I - Au + qu^2I)^{-1} \\ &= \left(\frac{q^2}{q^2u^2} - 1 \right)^{n/2} \left(\frac{1}{q^2u^2} - 1 \right)^{n/2} \det \left(I - A \frac{1}{qu} + I \frac{q}{(qu)^2} \right)^{-1} \\ &= (-1)^n A_X \left(\frac{1}{qu} \right). \quad \blacksquare \end{aligned}$$

Next we consider an *unramified* finite covering graph Y of our finite graph X . We assume that both X and Y are connected. A general discussion of covering spaces is given in Massey [10]. Of course, the case of graphs is simpler. We say that the graph Y is an unramified covering of the graph X iff we have a covering map $\pi: Y \rightarrow X$ which is an onto graph mapping (i.e., taking adjacent vertices to adjacent vertices) such that for every $x \in X$, and for every y in $\pi^{-1}(x)$, the collection of points adjacent to y in Y is mapped 1-1, onto the collection of points adjacent to $x \in X$. Suppose $\pi(\tilde{p}) = p$. One can show that every path in X starting at a point p has a unique lift to a path in Y starting at $\tilde{p}$.

If the graph X is $(q+1)$ -regular, then we often refer to the unramified covering graph Y as a $(q+1)$ -regular cover of X . One can also show that for an unramified covering, $\pi^{-1}(x)$ has the same number of elements for all x in X . If this number is d , we sometimes refer to Y as a d -sheeted covering of X in analogy to Riemann surface theory.

We can imitate Galois theory for these covering graphs. A d -sheeted covering is a *normal covering* if there are d automorphisms σ of Y preserving the projection map onto X , i.e., $\pi(\sigma(y)) = \pi(y)$, for all $y \in Y$. These automorphisms then form the *Galois group* $G(Y/X)$ of Y over X . The Galois group is the same as the *Covering group*.

COROLLARY 3. *If Y is a connected $(q+1)$ -regular covering of the $(q+1)$ -regular graph X , then $Z_X(u)^{-1}$ divides $Z_Y(u)^{-1}$.*

Proof. Note that if Y is connected, then X is connected. Also note that $r_X - 1 = n(q-1)/2$, where $n = |V(X)|$. So $r_X \leq r_Y$ and we can forget about the factor $(1-u^2)^{r_X-1}$ in $Z_X(u)^{-1}$. Next look at

$$\begin{aligned} (1-u^2)^{1-r_X} Z_X(u)^{-1} &= \det(I - A_X u + q u^2 I) \\ &= \prod_{\lambda \in \text{spec}(A_X)} (1 - \lambda u + q u^2) = p_X(u). \end{aligned}$$

In order to see that p_X divides p_Y , we need to see that

$$\text{spec}(A_X) \subset \text{spec}(A_Y).$$

To prove this, list the vertices of X as $v_1, \dots, v_n$, $n = |V(X)|$. Then list the vertices of Y according to which vertices of X they cover and which “sheet” of the covering they lie in:

$$\begin{aligned} &v_1^{(1)}, \dots, v_n^{(1)} \\ &v_1^{(2)}, \dots, v_n^{(2)} \\ &\dots \\ &v_1^{(m)}, \dots, v_n^{(m)}. \end{aligned}$$

Suppose that λ is an eigenvalue of A_X corresponding to eigenvector w . Create a new vector

$$z = \underbrace{({}^t w \quad {}^t w \quad \dots \quad {}^t w)}_m, \quad \text{where } {}^t v = \text{transpose of } v.$$

We claim that

$$A_Y z = \lambda z.$$

To see this, note that A_Y has block form

$$\begin{pmatrix} A_{11} & \dots & A_{1m} \\ \vdots & & \vdots \\ A_{m1} & \dots & A_{mm} \end{pmatrix}.$$

Here each row has the same number of 1's as in A_X and they are in the same location except for which block they lie in. We even have

$$\sum_{j=1}^m A_{ij} = A_X \quad \text{for every } i.$$

See the examples which follow. Hence

$$\begin{pmatrix} A_{11} & \cdots & A_{1m} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mm} \end{pmatrix} \begin{pmatrix} w \\ \vdots \\ w \end{pmatrix} = \begin{pmatrix} (A_{11} + \cdots + A_{1m}) w \\ \vdots \\ (A_{m1} + \cdots + A_{mm}) w \end{pmatrix} = \lambda \begin{pmatrix} w \\ \vdots \\ w \end{pmatrix}. \quad \blacksquare$$

Note. Corollary 3 actually holds for irregular graphs but a different proof will be required. See Corollary 1 of Theorem 2 in the next section.

EXAMPLE 1 (Two unramified quadratic coverings of the tetrahedron K_4). We build a graph Y_1 by drawing two copies of a spanning tree of the tetrahedron $X = K_4$ (dotted lines) and then drawing the rest of the edges of a two-fold cover Y_1 of X by connecting the remaining six edges of

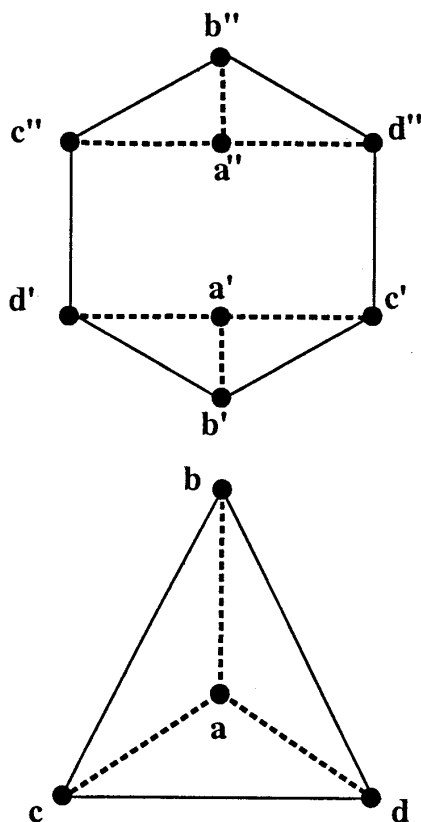


FIG. 2. A quadratic (double) covering of the tetrahedron K_4 . Dotted lines indicate spanning tree in K_4 and its covers.

Y_1 so that only two edges in Y_1 go between sheets of the cover. The construction is dependent on choices here (choice of tree and which edges go between). See Fig. 2.

We can easily compute the zeta functions of our examples using Ihara's Theorem 1. We find

$$Z_X(u)^{-1} = (1 - u^2)^2 p_X(u),$$

where

$$p_X(u) = (1 - u)(1 - 2u)(1 + u + 2u^2)^3$$

and

$$Z_{Y_1}(u)^{-1} = (1 - u^2)^4 (1 - u + 2u^2)(1 + u + 2u^2)(1 - u^2 + 4u^4) p_X(u).$$

We know by Corollary 3 that $Z_X(u)^{-1}$ divides $Z_{Y_1}(u)^{-1}$. Here we see it quite explicitly. If we recall what happens for Dedekind zeta functions of

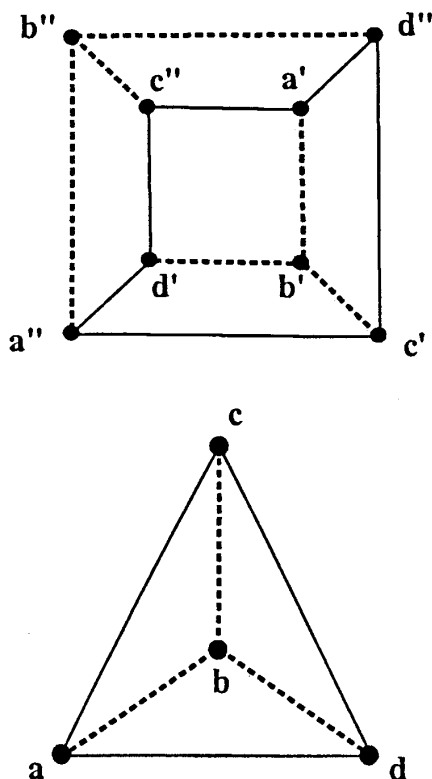


FIG. 3. The cube as a quadratic covering of the tetrahedron K_4 .

quadratic extensions of number fields, we might expect that $Z_{Y_1}(u)/Z_X(u)$ is an L -function associated to a non-trivial character of the Galois group $G(Y/X)$. In a later paper, we will discuss such L -functions and their factorization.

A second graph of this type Y_2 is constructed in the same way as Y_1 except that this time we make the connections on the two trees above so that all six edges change sheets. Our new covering turns out to be the cube.

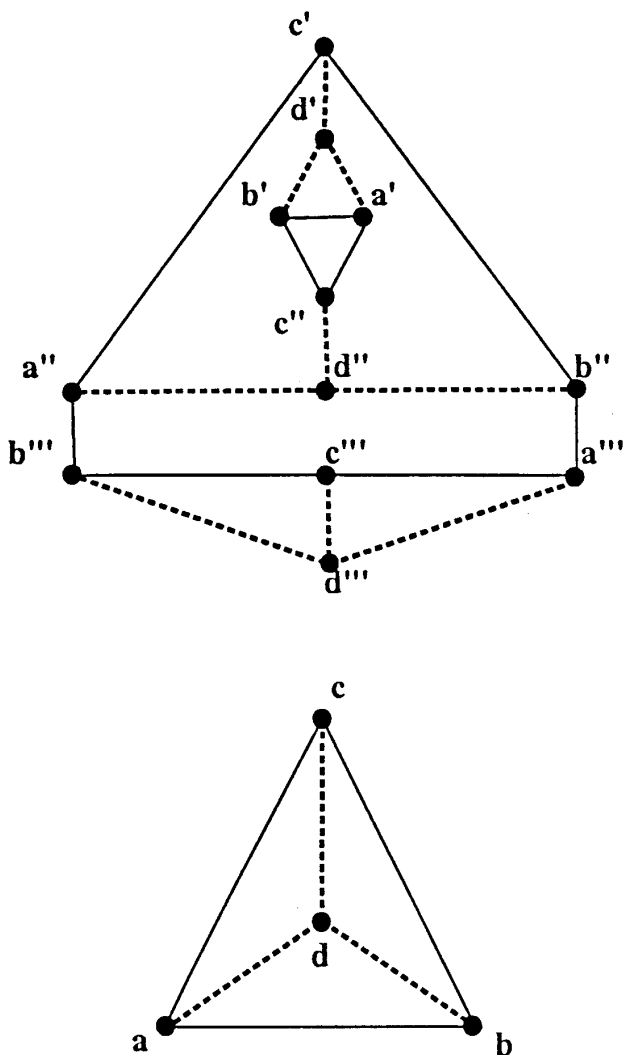


FIG. 4. Non-normal cubic covering Y_3 of K_4 .

EXAMPLE 2 (An unramified, non-normal, regular cubic cover of K_4). Now we start with a spanning tree in K_4 and make three copies of it. We make connections as indicated in the Fig. 4 to create the covering Y_3 of $X = K_4$.

We compute the inverse zeta function to be:

$$Z_{Y_3}(u)^{-1} = (1 - u^2)^6 (1 - u + 2u^2)^2 (1 + u + 2u^2)^2 \\ \times (1 - 3u^2 + 4u^4)(1 + u^2 + 4u^4) p_X(u),$$

where p_X is as before.

EXAMPLE 3 (A cyclic cubic normal cover of K_4). Once more create the covering Y_4 of X by making three copies of a spanning tree of X and then draw the extra nine edges as indicated. See Fig. 5.

We find the inverse zeta function to be

$$Z_{Y_4}(u)^{-1} = (1 - u^2)^6 (1 + u + 2u^2)^2 \\ \times (1 - u + u^2 - u^3 + 2u^4 - 4u^5 + 8u^6)^2 p_X(u),$$

where p_X is as before

Although distinct from each other, both $Z_{Y_1}(u)$ and $Z_{Y_2}(u)$ will turn out to be specializations of our universal rank 5 zeta function. Likewise both $Z_{Y_3}(u)$ and $Z_{Y_4}(u)$ will turn out to be specializations of our universal rank 7 zeta function.

2. COMBINATORIAL PROOF OF IHARA'S THEOREM FOR FINITE IRREGULAR GRAPHS

For an irregular graph X , the definition of the Ihara zeta function is the same as it was in the last section (see Eqs. (1.2) and (1.5)). But to generalize Theorem 1, we need to set

$$Q_X = Q = \text{an } n \times n \text{ diagonal matrix with } j\text{th diagonal entry } q_j.$$

Here $n = |V(E)|$, $q_j + 1 = \text{degree of } j\text{th vertex of } X$. We still have $r - 1 = \frac{1}{2} \text{Tr}(Q - I)$, where r is the rank of the fundamental group of X . However, note that the adjacency matrix A_X does not commute with Q_X . Thus, it is somewhat surprising that the following generalization of Theorem 1 holds.

THEOREM 2 (Ihara's theorem for irregular graphs. Bass [1]). *As usual $r = \text{rank fundamental group of } X$, and $r - 1 = \frac{1}{2} \text{Tr}(Q - I)$. Then the zeta function of X is given by*

$$Z_X(u)^{-1} = (1 - u^2)^{r-1} \det(I - Au + Qu^2).$$

Proof. We give a similar proof to that sketched in Sunada [14] for the regular case except that a simplification which we introduce in the discussion of "tail-counting" allows the extension to irregular graphs. The proof also begins in a similar way to that of Hashimoto [5] but then we diverge by not having to deal with bipartite graphs. Bass [1] proves Theorem 2 as an application of the formula for his multiedge zeta function.

Before beginning the proof, recall the definitions of Section 1—particularly, primitive closed path with no backtracking and no tails.

First take a logarithm of formula (1.2)

$$\log Z_X(u) = - \sum_{[C]} \log(1 - u^{v(C)}) = \sum_{[C]} \sum_{j \geq 1} \frac{1}{j} u^{v(C)j},$$

where the sum over $[C]$ is over equivalence classes of primitive, closed paths in X with no backtracking and no tails as defined before formula (1.2). Thus,

$$u \frac{d}{du} \log Z_X(u) = \sum_{j \geq 1} \sum_{d \geq 1} d \sum_{\substack{[C] \\ v(C)=d}} u^{v(C)j}$$

Now there are d elements in $[C]$. So we can drop the $[]$ and have

$$u \frac{d}{du} \log Z_X(u) = \sum_{j \geq 1} \sum_{d \geq 1} \sum_{\substack{C \\ v(C)=d}} u^{v(C)j} = \sum_{j \geq 1} \sum_C u^{v(C)j}.$$

The sum over j and primitive C combines to a sum over all closed paths C with no backtracking and no tails (no longer primitive) yielding

$$u \frac{d}{du} \log Z_X(u) = \sum_{m \geq 1} N_m u^m, \quad (2.1)$$

where

$$N_m = \begin{cases} \text{number of closed paths } C \text{ in } X \text{ of length } m \\ \text{with no backtracking and no tails.} \end{cases} \quad (2.2)$$

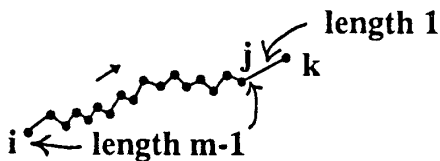


FIG. 6. Counting the no backtracking paths of length m from i to k .

Now to get to the other side of the formula in Theorem 2, for $m \geq 1$, we introduce an $n \times n$ matrix A_m whose i, j entry is given by

$$(A_m)_{i,j} = \begin{cases} \text{number of paths } C \text{ in } X \text{ of length } m \\ \text{with no backtracking} \\ \text{starting at } i \text{ and ending at } j. \end{cases} \quad (2.3)$$

Here i, j run through the vertices of X . We also define $A_0 = I$. Note that $A_1 = A$.

LEMMA 1 (Recursions for the matrices A_m). *We have*

$$A_2 = A^2 - (Q + I)$$

and for $m \geq 3$,

$$A_m = A_{m-1}A - A_{m-2}Q.$$

Proof. Note that if $i \neq j$, we have

$$(A_2)_{i,j} = (A^2)_{i,j},$$

since both sides are equal to the number of paths from i to j of length 2.

However, if $i = j$, we have

$$(A_2)_{i,i} = 0,$$

as there are no paths from i to i of length 2 without backtracking. But

$$(A^2)_{i,i} = q_i + 1 = \text{degree of the } i\text{th vertex.}$$

This gives the formula stated in Lemma 1 for $m = 2$.

Now, assume $m \geq 3$ and look at Fig. 6.

We want to count the no backtracking paths of length m from i to k as above. The sum

$$\sum_j (A_{m-1})_{i,j} A_{j,k}$$

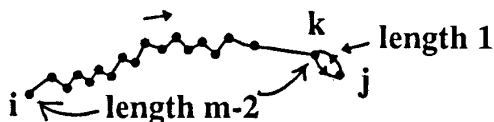


FIG. 7. Backtracking paths of length m from i to k included in $\sum_j (A_{m-1})_{i,j} A_{j,k}$.

counts these paths plus additional paths with backtracking of the sort pictured in Fig. 7. How many such extra paths are there? Since the length $m-1$ path from i to k to j has no backtracking, j is any of q_k adjacent vertices to k with the remaining adjacent vertex to k being ineligible. Thus the number of excess paths is

$$(A_{m-2})_{i,k} q_k = (A_{m-2} Q)_{i,k}.$$

This implies the formula in Lemma 1 for $m \geq 3$. ■

From Lemma 1 we obtain

$$\left(\sum_{m \geq 0} A_m u^m \right) (I - Au + Qu^2) = (1 - u^2) I. \quad (2.4)$$

Now multiply (2.4) by $1/(1 - u^2) = \sum_{j \geq 0} u^{2j}$ to obtain

$$I = \left(\sum_{k \geq 0} A_k u^k \right) \left(\sum_{j \geq 0} u^{2j} \right) (I - Au + Qu^2).$$

This becomes

$$I = \left(\sum_{m \geq 0} \sum_{j=0}^{[m/2]} A_{m-2j} u^m \right) (I - Au + Qu^2). \quad (2.5)$$

Here $[x]$ = greatest integer $\leq x$. Both (2.4) and (2.5) will be important in the sequel.

Our next problem is to go from the A_m 's to the N_m 's. This means we must count tails. Define for $m \geq 1$,

$$t_m = \begin{cases} \text{the number of closed paths} \\ \text{of length } m \text{ with tails} \\ \text{and having no backtracking.} \end{cases} \quad (2.6)$$

Note that there are no paths of length 1 or 2 of this type so that $t_1 = t_2 = 0$.

LEMMA 2. For $m \geq 3$, $t_m = \text{Tr}[(Q - I) A_{m-2}] + t_{m-2}$.

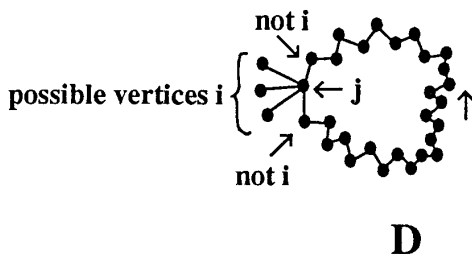


FIG. 8. The case where D has no tail. In order to avoid backtracking, i cannot be one of the two vertices on D one step away from j .

Proof. Note that a tail must occur at the beginning of the path or there will be backtracking. Now

$$\begin{aligned}
 t_m &= \sum_i \# \left\{ \begin{array}{l} \text{closed paths } C \text{ with tails, no} \\ \text{backtracking, } v(C) = m, \text{ starting at } i \end{array} \right\} \\
 &= \sum_{\substack{i, j \\ i \text{ adjacent to } j}} \# \left\{ \begin{array}{l} \text{closed paths } C \text{ with tails, no} \\ \text{backtracking, } v(C) = m, \text{ starting} \\ \text{at } i, \text{ going to } j \text{ at first step} \end{array} \right\} \\
 &= \sum_j \sum_{i \text{ adjacent to } j} \# \left\{ \begin{array}{l} \text{closed paths } C \text{ with tails, no} \\ \text{backtracking, } v(C) = m, \text{ starting} \\ \text{at } i, \text{ going to } j \text{ at first step} \end{array} \right\}.
 \end{aligned}$$

Such a path C goes from i to j and then over a path D from j to j of length $m-2$ and then from j to i . There are two kinds of paths D of length $m-2$ from j to j —those without tails and those with tails. Any path D of length $m-2$ from j to j potentially contributes. If the path D does not have a tail, in order to avoid backtracking i must be one of the q_j-1 vertices which are adjacent to j and not one step away from j on D . See Fig. 8.

If the path D does have a tail, then vertex i must be one of the q_j vertices which are adjacent to j and not one step away from j on D . See Fig. 9.

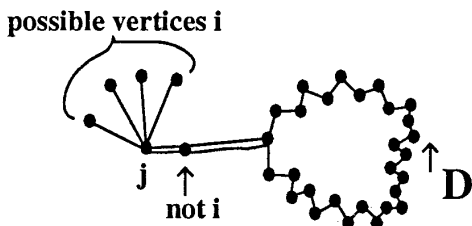


FIG. 9. If D does have a tail, i cannot be the one vertex on D one step away from j .

Now take the inner sum over i and collapse it. We have two contributions:

$$\begin{aligned}
 \sum_i &= \sum_{i \text{ adjacent to } j} \# \left\{ \begin{array}{l} \text{closed paths } C \text{ with tails, no} \\ \text{backtracking, } v(C) = m, \text{ starting} \\ \text{at } i, \text{ going to } j \text{ at first step} \end{array} \right\} \\
 &= (q_j - 1) \# \left\{ \begin{array}{l} \text{closed paths } D \text{ without} \\ \text{tails, no backtracking,} \\ v(D) = m - 2, \text{ starting at } j \end{array} \right\} \\
 &\quad + q_j \# \left\{ \begin{array}{l} \text{closed paths } D' \text{ with} \\ \text{tails, no backtracking,} \\ v(D') = m - 2, \text{ starting at } j \end{array} \right\}.
 \end{aligned}$$

Rewrite this as

$$\begin{aligned}
 \sum_i &= (q_j - 1) \# \left\{ \begin{array}{l} \text{closed paths } D, \text{ no} \\ \text{backtracking, } v(D) = m - 2, \\ \text{starting at } j \end{array} \right\} \\
 &\quad + \# \left\{ \begin{array}{l} \text{closed paths } D' \text{ with} \\ \text{tails, no backtracking,} \\ v(D') = m - 2, \text{ starting at } j \end{array} \right\}
 \end{aligned}$$

It follows that

$$\begin{aligned}
 t_m &= \sum_j \sum_i \\
 &= \sum_j (q_j - 1) \# \left\{ \begin{array}{l} \text{closed paths } D, \text{ no} \\ \text{backtracking, } v(D) = m - 2 \\ \text{starting at } j \end{array} \right\} \\
 &\quad + \sum_j \# \left\{ \begin{array}{l} \text{closed paths } D' \text{ with} \\ \text{tails, no backtracking,} \\ v(D') = m - 2, \text{ starting at } j \end{array} \right\}.
 \end{aligned}$$

This is exactly the formula stated in Lemma 2:

$$t_m = \text{Tr}((Q - I) A_{m-2}) + t_{m-2}. \quad \blacksquare$$

This allows us to derive the following important *relation between the A 's and N 's*:

$$N_m = \text{Tr} \left(A_m - (Q - I) \sum_{j=1}^{[(m-1)/2]} A_{m-2j} \right) \quad \text{for } m \geq 3. \quad (2.7)$$

Also $N_2 = N_1 = \text{Tr } A_2 = \text{Tr } A_1 = 0$. To prove (2.7), use Lemma 2 and induction to see that

$$t_m = \text{Tr}[(Q - I) A_{m-2}] + t_{m-2} = \text{Tr} \left((Q - I) \sum_{j=1}^{[(m-1)/2]} A_{m-2j} \right).$$

The induction stops before you hit A_0 , since $t_2 = t_1 = 0$. Formula (2.7) follows since $N_m = \text{Tr}(A_m) - t_m$.

Next define a matrix N_m^* for $m \geq 0$ by

$$N_m^* = A_m - (Q - I) \sum_{j=1}^{[m/2]} A_{m-2j} = QA_m - (Q - I) \sum_{j=0}^{[m/2]} A_{m-2j}. \quad (2.8)$$

Note the change from (2.7) in the upper bound on the sum! We have then by (2.7) for $m \geq 1$,

$$\text{Tr}(N_m^*) = N_m - \begin{cases} 0, & m \text{ odd,} \\ \text{Tr}(Q - I), & m \text{ even.} \end{cases} \quad (2.9)$$

It follows that

$$\begin{aligned} & \left(\sum_{m \geq 0} N_m^* u^m \right) (I - Au + Qu^2) \\ &= \left(Q \sum_{m \geq 0} A_m u^m - (Q - I) \sum_{m \geq 0} \sum_{j=0}^{[m/2]} A_{m-2j} u^m \right) (I - Au + Qu^2). \end{aligned}$$

Now we use (2.4) and (2.5) to see that this is $(1 - u^2) Q - (Q - I) I = I - Qu^2$. Since $N_0^* = A_0 = I$, we have

$$\left(\sum_{m \geq 1} N_m^* u^m \right) (I - Au + Qu^2) = I - Qu^2 - I(I - Au + Qu^2) = Au - 2Qu^2.$$

So we have proved

$$\left(\sum_{m \geq 1} N_m^* u^m \right) = (Au - 2Qu^2)(I - Au + Qu^2)^{-1}. \quad (2.10)$$

The right-hand side of (2.10) is formally u times a logarithmic derivative but A and Q do not commute. The trace cures that problem.

LEMMA 3. *Let $f(u)$ be a square matrix and assume $f(0)=0$. Then*

$$\mathrm{Tr} \left(-\frac{d}{du} \log(I - f(u)) \right) = \mathrm{Tr}(f'(u)(I - f(u))^{-1}).$$

In our application, $f(u) = Au - Qu^2$.

Proof. We have

$$-\log(I - f(u)) = \sum_{n \geq 1} \frac{1}{n} f(u)^n, \quad \frac{d}{du} (f(u))^n = \sum_{j=0}^{n-1} f(u)^j f'(u) f(u)^{n-j-1}.$$

It follows that

$$\begin{aligned} \mathrm{Tr} \left(\frac{d}{du} (f(u))^n \right) &= \mathrm{Tr} \left(\sum_{j=0}^{n-1} f(u)^j f'(u) f(u)^{n-j-1} \right) \\ &= \mathrm{Tr} \left(\sum_{j=0}^{n-1} f'(u) f(u)^{n-1} \right) \\ &= n \mathrm{Tr}(f'(u) f(u)^{n-1}). \end{aligned}$$

Therefore,

$$\begin{aligned} \mathrm{Tr} \left(-\frac{d}{du} \log(I - f(u)) \right) &= \sum_{n \geq 1} \mathrm{Tr}(f'(u) f(u)^{n-1}) \\ &= \mathrm{Tr} \left(f'(u) \sum_{n \geq 1} f(u)^{n-1} \right) \\ &= \mathrm{Tr}(f'(u)(I - f(u))^{-1}). \quad \blacksquare \end{aligned}$$

It follows from Lemma 3 that

$$\mathrm{Tr} \left(\sum_{m \geq 1} N_m^* u^m \right) = \mathrm{Tr} \left(-u \frac{d}{du} \log(I - Au + Qu^2) \right). \quad (2.11)$$

By (2.9), the left-hand side of (2.11) is

$$\sum_{m \geq 1} N_m u^m - \mathrm{Tr}(Q - I) \left(\sum_{\substack{m \text{ even} \\ m \geq 2}} u^m \right) = \sum_{m \geq 1} N_m u^m - \mathrm{Tr}(Q - I) \frac{u^2}{1 - u^2}.$$

So by (2.1) and (2.11), we have

$$\begin{aligned}
 u \frac{d}{du} \log Z_X(u) &= \sum_{m \geq 1} N_m u^m \\
 &= \operatorname{Tr} \left(-u \frac{d}{du} \log(I - Au + Qu^2) \right) + \operatorname{Tr}(Q - I) \frac{u^2}{1 - u^2} \\
 &= \operatorname{Tr} \left(-u \frac{d}{du} \log(I - Au + Qu^2) \right) - u \frac{d}{du} \log((1 - u^2)^{\operatorname{Tr}(Q - I)/2}).
 \end{aligned}$$

Our functions are 0 at $u=0$ and, thus, we can integrate to get

$$-\log Z_X(u) = \operatorname{Tr}(\log(I - Au + Qu^2)) + \log((1 - u^2)^{\operatorname{Tr}(Q - I)/2}).$$

Since

$$\exp(\operatorname{Tr}(\log(I - Au + Qu^2))) = \det(I - Au + Qu^2),$$

this proves Theorem 2. ■

Next we generalize Corollary 3 of Section 1 to unramified, possibly irregular, non-normal connected covers Y of X .

COROLLARY 1 (Zeta functions of unramified covers). *For every unramified possibly irregular and possibly non-normal connected covering Y of X the inverse zeta function $Z_X^{-1}(u)$ divides $Z_Y^{-1}(u)$.*

Proof. We assume that Y is a d -fold cover of X . First note that $r_Y \geq r_X$. We will write A for the adjacency matrix of X , Q for the diagonal matrix of degrees as defined at the beginning of this section, and I for the corresponding identity matrix. We will also write $\tilde{A}$ for the adjacency matrix of Y , $\tilde{Q}$ for the diagonal matrix of degrees of the vertices of Y , and $\tilde{I}$ for the corresponding identity matrix. We can put $\tilde{A}$ in block form with

$$\tilde{A} = (A_{ij})_{1 \leq i, j \leq d} \text{ such that for each } i, \quad \sum_j A_{ij} = A.$$

And $\tilde{Q}$ has block diagonal form with d copies of Q down the diagonal. Similarly $\tilde{I}$ has block diagonal form with d copies of I down the diagonal.

Look at

$$\tilde{I} - \tilde{A}u + \tilde{Q}u^2.$$

For the determinant add the right $d-1$ block columns to the first block column. Since $I+Qu^2$ only appears once per block row, the new first column is

$$\begin{pmatrix} I - Au + Qu^2 \\ \vdots \\ I - Au + Qu^2 \end{pmatrix}.$$

Now subtract the first block row from all remaining rows; the first block column becomes

$$\begin{pmatrix} I - Au + Qu^2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This implies that $\det(I - Au + Qu^2)$ divides $\det(\tilde{I} - \tilde{A}u + \tilde{Q}u^2)$. ■

Now let us look at some examples of zeta functions of coverings of irregular graphs.

EXAMPLE 4 (A Klein 4-group covering of three quadratic coverings of a subgraph X of K_4). Here we consider three quadratic coverings Y_1, Y_2, Y_3 of a subgraph X of K_4 pictured in Fig. 10, along with a quartic cover K sitting above everything. Here Y_2 and Y_3 are isomorphic as graphs.

We compute the zeta functions as follows:

- (1) the zeta function of X is

$$Z_X(u)^{-1} = (1 - u^2) p_1(u),$$

where

$$p_1(u) = (1 - u)(1 + u^2)(1 + u + 2u^2)(1 - u^2 - 2u^3);$$

- (2) the zeta function of Y_1 is

$$Z_{Y_1}(u)^{-1} = (1 - u^2)^2 p_2(u),$$

where

$$p_2(u) = p_1(u)(1 + u)(1 + u^2)(1 - u + 2u^2)(1 - u^2 + 2u^3);$$

- (3) the zeta function of the isomorphic covers Y_2 and Y_3 is

$$Z_{Y_3}(u)^{-1} = Z_{Y_2}(u)^{-1} = (1 - u^2)^2 q_2(u),$$

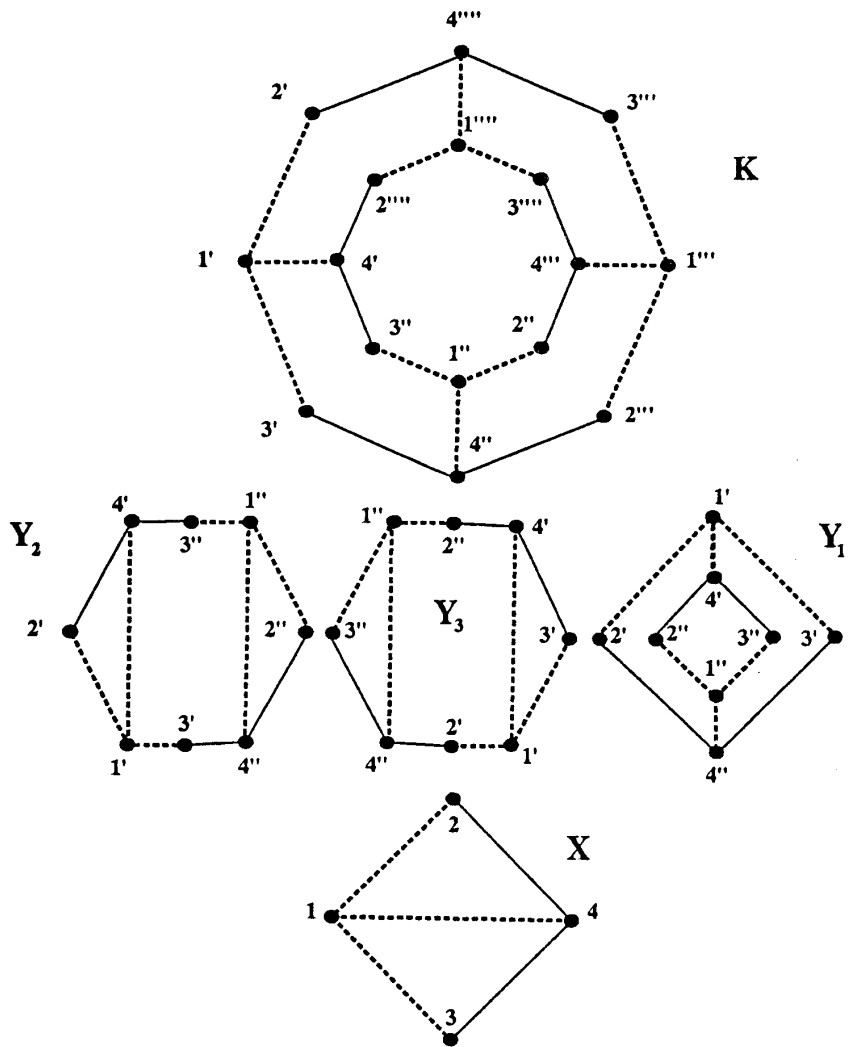


FIG. 10. A Klein 4-group cover K of 3 quadratic covers Y_1, Y_2, Y_3 of a subgraph X of K_4 .

where

$$q_2(u) = p_1(u)(1 - u + u^2 - u^3 + 2u^4)(1 + u + u^2 + u^3 + 2u^4);$$

(4) the zeta function of K is

$$Z_K(u)^{-1} = (1 - u^2)^4 p_4(u),$$

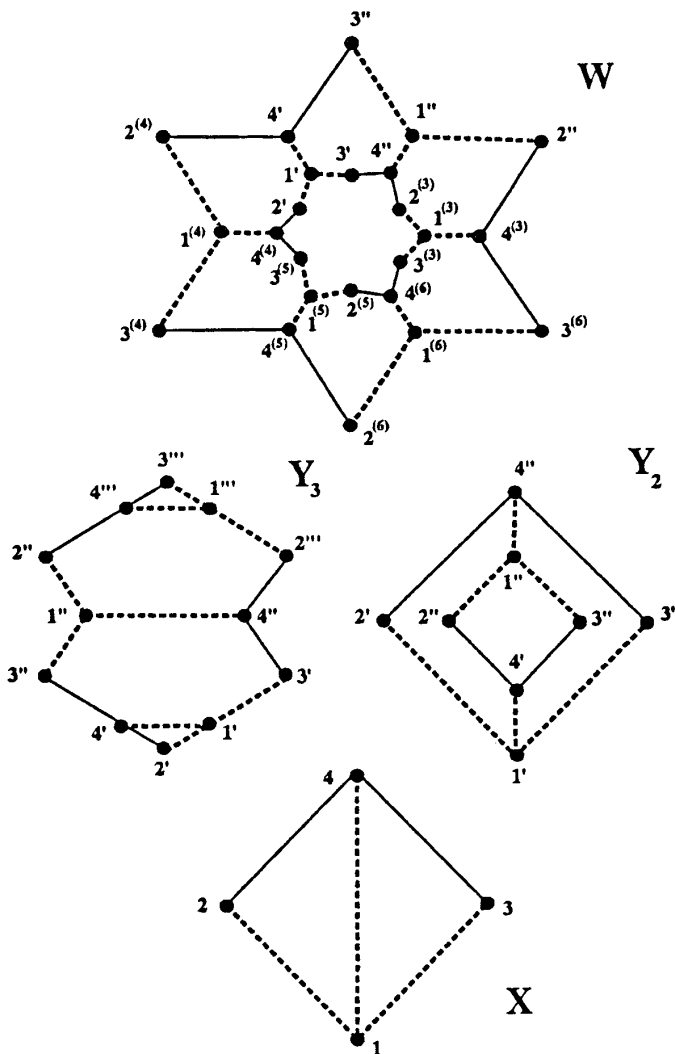


FIG. 11. A sextic S_3 covering W of a quadratic cover Y_2 and a cubic cover Y_3 of a subgraph X of K_4 .

where

$$p_4(u) = p_2(u)(1 - u + u^2 - u^3 + 2u^4)^2 (1 + u + u^2 + u^3 + 2u^4)^2.$$

Just as would be the case for zeta functions of number fields, we find that

$$(p_1)^2 p_4 = p_2(q_2)^2$$

and

$$Z_X(u)^2 Z_K(u) = Z_{Y_1}(u) Z_{Y_2}(u) Z_{Y_3}(u).$$

EXAMPLE 5 (A normal sextic covering W containing a quadratic and cubic covering of a subgraph of K_4 . The Galois group of W is S_3). The sextic normal covering W of X pictured in Fig. 11 has the symmetric group on three elements as its covering group $G(W/X)$. The subcover Y_2 is quadratic and the subcover Y_3 is cubic (non-normal). We find that just as in the previous example, we have a factorization of zeta functions similar to that which would occur for those of number fields—an indication that there must be a similar theory of Artin L -functions.

We compute the zeta functions to be:

- (1) the zeta function of X is

$$Z_X(u)^{-1} = (1 - u^2) p_1(u),$$

where

$$p_1(u) = (1 - u)(1 + u^2)(1 + u + 2u^2)(1 - u^2 - 2u^3);$$

- (2) the zeta function of the quadratic cover Y_2 is

$$Z_{Y_2}(u)^{-1} = (1 - u^2)^2 p_2(u),$$

where

$$p_2(u) = p_1(u)(1 + u)(1 + u^2)(1 - u + 2u^2)(1 - u^2 + 2u^3);$$

- (3) the zeta function of the cubic cover Y_3 is

$$Z_{Y_3}(u)^{-1} = (1 - u^2)^3 p_3(u),$$

where

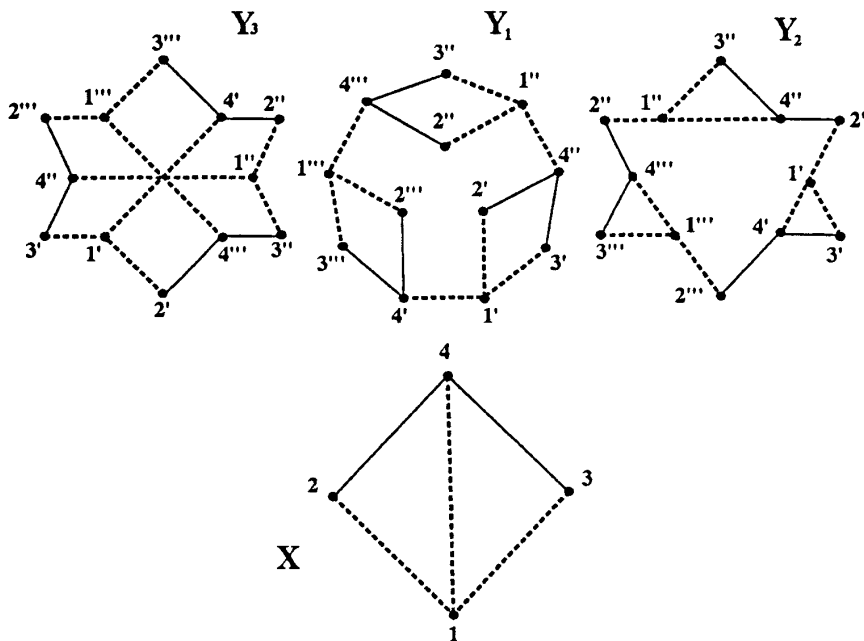
$$\begin{aligned} p_3(u) &= p_1(u)(1 - u - u^3 + 2u^4)(1 - u + 2u^2 - u^3 + 2u^4) \\ &\quad \times (1 + u + u^3 + 2u^4)(1 + u + 2u^2 + u^3 + 2u^4); \end{aligned}$$

- (4) the zeta function of the sextic cover W is

$$Z_W(u)^{-1} = (1 - u^2)^6 p_6(u),$$

where

$$\begin{aligned} p_6(u) &= p_3(u)(1 + u)(1 + u^2)(1 - u + 2u^2)(1 - u^2 + 2u^3) \\ &\quad \times (1 - u - u^3 + 2u^4)(1 - u + 2u^2 - u^3 + 2u^4) \\ &\quad \times (1 + u + u^3 + 2u^4)(1 + u + 2u^2 + u^3 + 2u^4). \end{aligned}$$

FIG. 12. The three types of normal cubic cover of X .

Just as would be the case for zeta functions of number fields, we find that

$$(p_1)^2 p_6 = p_2(p_3)^2$$

and

$$Z_X(u)^2 Z_W(u) = Z_{Y_2}(u) Z_{Y_3}(u)^2.$$

EXAMPLE 6 (The three types of normal cubic covers of a subgraph of K_4). Here we consider the three types of normal cubic covers of the same subgraph X of K_4 considered in the previous two examples. The graphs are pictured in Fig. 12.

We find that the zeta functions of these graphs are as follows:

- (1) the zeta function of X is

$$Z_X(u)^{-1} = (1 - u^2) p(u),$$

where

$$p(u) = (1 - u)(1 + u^2)(1 + u + 2u^2)(1 - u^2 - 2u^3);$$

(2) the zeta function of Y_1 is

$$Z_{Y_1}(u)^{-1} = (1 - u^2) p_1(u),$$

where

$$p_1(u) = p(u)(1 + u^2)^2 (1 + 2u^3 - u^4 + 4u^6)^2;$$

(3) the zeta function of Y_2 is

$$Z_{Y_2}(u)^{-1} = (1 - u^2) p_2(u),$$

where

$$p_2(u) = p(u)(1 + u^2 - u^3 + 2u^4 - u^5 + 3u^6 + 4u^8)^2;$$

(4) the zeta function of Y_3 is

$$Z_{Y_3}(u)^{-1} = (1 - u^2) p_3(u),$$

where

$$p_3(u) = p(u)(1 - u + 2u^2 - u^3 + 2u^4)^2 (1 + u + u^3 + 2u^4)^2.$$

3. THE EDGE ZETA FUNCTION

To motivate this section, note that by Theorem 2, the degree of the polynomial $Z_X(u)^{-1}$ is the same as

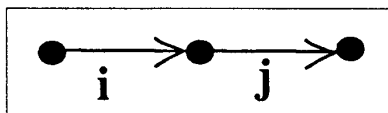
$$2r - 2 + 2|V| = 2(|E| - |V|) + 2|V| = 2|E|.$$

This suggests that if you take twice as many variables as edges and form a particularly nice matrix of that size, you might hope to get $Z_X(u)^{-1}$ out of it by specializing all of the variables to be equal to u . Also, you can use two variables for each edge, one attached to each possible direction, to keep track of directed paths on the graph.

For any finite graph X , we can define a $2|E| \times 2|E|$ matrix M called the directed edge matrix. This is done by taking $a_1, \dots, a_{|E|}$ to correspond to the directed edges of X . It does not matter how you choose directions. Then b_i corresponds to the opposite direction from a_i .

DEFINITION. The *directed edge matrix* M has as its ij entry

$$M_{ij} = \begin{cases} 1, & \text{if edge } i \text{ feeds into edge } j \\ & \text{to form a no backtrack path;} \\ 0, & \text{otherwise.} \end{cases} \quad (3.1)$$

FIG. 13. Paths ij with $M_{ij} = 1$.

The picture of the paths ij necessary for M_{ij} to be 1 is in Fig. 13.

Before constructing the multi-variable zeta function, let us look at an example.

EXAMPLE 7 (Our favorite subgraph of K_4). We will work out the directed edge matrix M for the graph in Fig. 14. The matrix M for the graph in Fig. 14 is

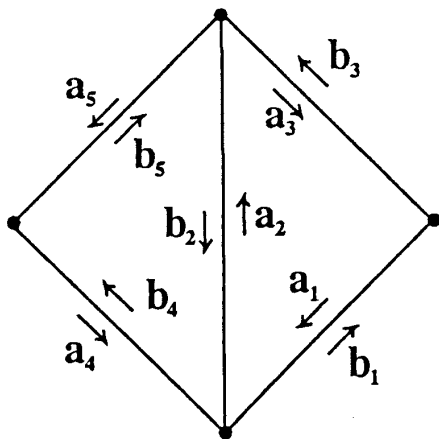
	a_1	a_2	a_3	a_4	a_5	b_1	b_2	b_3	b_4	b_5
a_1		1							1	
a_2			1		1					
a_3	1									
a_4		1				1				
a_5				1						
b_1								1		
b_2						1			1	
b_3					1		1			
b_4										1
b_5			1				1			

The matrix M has the following properties.

LEMMA 4 (Properties of the directed edge matrix). *Write*

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where each block is an $|E| \times |E|$ matrix, and we are assuming that the directed edges have been labeled as in the example above. Then we have

FIG. 14. Our favorite subgraph of K_4 .

$$(1) \quad {}^tB = B \text{ and } {}^tC = C.$$

$$(2) \quad D = {}^tA.$$

$$(3) \quad m_{a_i b_i} = 0 = m_{b_i a_i}.$$

Proof. (1) If the directed edge a_i feeds into b_j , then a_j feeds into b_i . Likewise if b_i feeds into a_j , then b_j feeds into a_i .

(2) If a_i feeds into a_j , then b_j feeds into b_i .

(3) We have $m_{a_i b_i} = 1$ only if the path formed by $a_i b_i$ has no backtracking. But this is not the case. ■

DEFINITION. Let U be the diagonal matrix

$$U = \text{diag}(u_{a_1}, \dots, u_{a_{|E|}}, u_{b_1}, \dots, u_{b_{|E|}}). \quad (3.2)$$

Any directed path on X is represented by a sequence of numbers; e.g., as in Fig. 15, the path 1354 corresponds to $b_1 b_3 a_5 a_4$.

DEFINITION. The *edge zeta function* associated to the set $\mathbf{E}$ of all directed edges of X is defined as a function of $u_j \in \mathbb{C}$ (sufficiently near 0), $j = 1, \dots, |\mathbf{E}|$, by

$$\zeta_X(u) = \prod_{\substack{[C] \text{ classes of primitive} \\ \text{cycles in } X \text{ with no} \\ \text{backtracking and no trails}}} (1 - g(C))^{-1}, \quad (3.3)$$

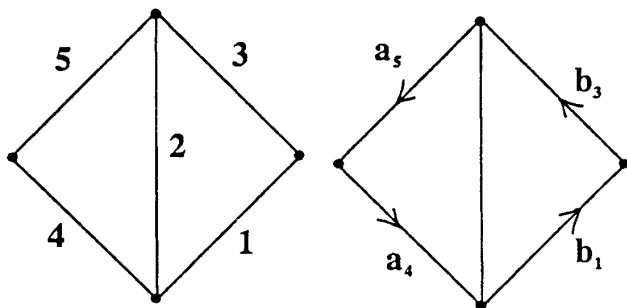


FIG. 15. A path in the graph corresponding to $b_1 b_3 a_5 a_4$.

where

$$g(C) = \prod_{\mathbf{e} \text{ edge in } C} u_{\mathbf{e}},$$

and edges in the product occur according to multiplicity. The product over equivalence classes $[C]$ is as in (1.2). The product defining $g(C)$ is over directed edges $\mathbf{e}$ in C . It does not depend on the choice of C in its class.

EXAMPLE. Using the numbering of Figs. 14 and 15, we have

$$\begin{array}{cccc} 132 & 132 & 452 & 452 \\ 132 & 452 & 452 & 132 \end{array}$$

are primitive equivalent paths with the same

$$g(C) = u_{b_1}^2 u_{b_2}^4 u_{b_3}^2 u_{b_4}^2 u_{b_5}^2.$$

PROPOSITION 1 (Specializing variables to 0 gives zeta functions of subgraphs). Suppose that X is a graph with edge set $E = E(X)$. Let F be a subset of E . Let $\mathbf{F}$ consist of all directed edges coming from F . Suppose W is the graph obtained from X by erasing all edges in F . Then

$$\zeta_X(u)|_{u_{\mathbf{e}}=0, \forall \mathbf{e} \in \mathbf{F}} = \zeta_W(u).$$

Here we are just setting all the variables corresponding to edges in F equal to 0.

Proof. This is easy since when we set all the variables equal to 0 corresponding to edges in F , and we look at the product over $[C]$ in (3.3), any term will become 1 if C contains an edge from F . Thus the paths must stay in W for the term not to equal 1. ■

PROPOSITION 2 (The edge zeta function specializes to the Ihara zeta function when all variables are set equal).

$$\zeta_X(u) \big|_{u_j = u, \text{ all } j} = Z_X(u).$$

Proof. This is clear from (3.3) and (1.2). ■

An example showing how these propositions and the following theorem work for the graph of Fig. 14 is included at the end of this section. Now we prove the analogue of Ihara's theorem for the multivariable edge zeta function.

THEOREM 3 (Generalization of Ihara's theorem for the edge zeta function. Bass [1]). *With the definitions (3.1) and (3.3), we have*

$$\zeta_X(u)^{-1} = \det(I - MU) = \det(I - UM).$$

Notes. (1) This theorem is surprisingly simpler for the multivariable zeta function than were Theorems 1 and 2. We will find that the proof is simpler too! However, it is not obvious how to see that if we specialize all variables to be equal, the $\det(I - UM)$ factors into $(1 - u^2)^{r-1} \det(I - Au + Qu^2)$ —something we know must happen by Theorem 2 of the last section. But this how Bass [1] proves Theorem 2.

(2) You can use Theorem 3 to get another proof of Proposition 1 by expansion by minors along the rows wiped out by deleting edges.

Proof. As usual, start by taking logs,

$$\log \zeta_X(u)^{-1} = \sum_{[C]} \sum_{j \geq 1} \frac{1}{j} g(C)^j.$$

where the sum over $[C]$ is over classes of primitive cycles with no backtracking and no tails.

This implies that if $\mathbf{E}$ denotes the set of $2|E|$ directed edges in X , we have

$$\sum_{\mathbf{e} \in \mathbf{E}} u_{\mathbf{e}} \frac{\partial}{\partial u_{\mathbf{e}}} \log \zeta_X^{-1} = \sum_{m \geq 1} \frac{1}{m} \sum_{\substack{C \text{ cycle length } m \\ \text{primitive, no backtrack} \\ \text{no tails}}} \sum_{j \geq 1} \frac{1}{j} \sum_{\mathbf{e}} u_{\mathbf{e}} \frac{\partial}{\partial u_{\mathbf{e}}} g(C)^j.$$

Here we have used the fact that a primitive path of length m has m equivalent paths of length m .

Next we use the formula

$$\sum_{j=1}^J u_j \frac{\partial}{\partial u_j} (u_1^{r_1} \cdots u_J^{r_J}) = \left(\sum_{j=1}^J r_j \right) (u_1^{r_1} \cdots u_J^{r_J}). \quad (3.4)$$

Collect terms in the u 's and obtain

$$g(C)^j = \prod_{\mathbf{e} \in C} u_{\mathbf{e}}^{jr_{\mathbf{e}}}, \quad \text{where} \quad \sum_{\mathbf{e} \in C} r_{\mathbf{e}} = m.$$

The total degree is jm and the jm 's cancel.

We find that

$$\sum_{\mathbf{e}} u_{\mathbf{e}} \frac{\partial}{\partial u_{\mathbf{e}}} \log \zeta_X^{-1} = \sum_{m \geq 1} \sum_{\substack{C \text{ cycle, } v(C)=m \\ \text{no backtrack} \\ \text{no tails}}} \prod_{\mathbf{e} \in C} u_{\mathbf{e}}. \quad (3.5)$$

Here we have dropped the words “primitive” and “up to equivalence” in the sum over C .

DEFINITION. We define the *multivariable analogue* of N_m in (2.2) as

$$N_m(u) = \sum_{\substack{C \text{ cycle, } v(C)=m \\ \text{no backtrack} \\ \text{no tails}}} \prod_{\mathbf{e} \in C} u_{\mathbf{e}}. \quad (3.6)$$

We need the following formula for $N_m(u)$. This is much simpler than the analogous result for N_m in (2.7) plus the recursions for the A_m 's in Lemma 1 of Section 2.

LEMMA 5 (Formula relating $N_m(u)$ and the matrices M and U). *We have*

$$N_m(u) = \text{Tr}[(MU)^m] = \text{Tr}[(UM)^m].$$

Proof. Consider the Example of Fig. 14 again. Look for the $(1, 1)$ entry in $UMUMUM$. You only find a non-zero contribution of the form

$$u_{a_1} u_{a_2} u_{a_3} = (UM)_{1,2} (UM)_{2,3} (UM)_{3,1}$$

if path a_1 feeds into a_2 and path a_2 feeds into a_3 and path a_3 feeds into a_1 . And it would not be in the trace unless the path represented were closed. So such a term exists says we have a path which must be tail-less, no backtracking, length 3, and closed. Generalizing this argument, we have

$$((UM)^m)_{\mathbf{i}, \mathbf{j}} = \# \left\{ \begin{array}{l} \text{paths } C \text{ of length } m, \text{ starting at edge } \mathbf{i} \text{ and} \\ \text{ending with an edge which feeds into edge } \mathbf{j}, \\ \text{having no tails, and no backtracking} \end{array} \right\} \prod_{\mathbf{e} \in C} u_{\mathbf{e}}.$$

Taking the trace gives a proof of the lemma. ■

We return to the proof of Theorem 3. From (3.5), (3.6), and Lemma 5 we have

$$\sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} \log \zeta_X^{-1} = \sum_{m \geq 1} N_m(u) = \sum_{m \geq 1} \text{Tr}[(MU)^m]. \quad (3.7)$$

We need a lemma.

LEMMA 6. *Suppose that the matrix U is a function of x . Then we have*

$$\frac{d}{dx} (\text{Tr}[(MU)]^m) = \text{Tr} \left(m(MU)^{m-1} \frac{d}{dx} (MU) \right).$$

Proof. Use the same sort of argument that we gave for Lemma 3 in the last section. ■

Now we need to work on the “other” side of Theorem 3. Using Lemma 6 and

$$\log(\det B) = \text{Tr}(\log B),$$

we have

$$\begin{aligned} \sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} \log \det(I - MU)^{-1} &= \sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} \text{Tr} \left(\sum_{m \geq 1} \frac{1}{m} (MU)^m \right) \\ &= \sum_{m \geq 1} \sum_{\mathfrak{e}} u_{\mathfrak{e}} \text{Tr} \left((MU)^{m-1} \frac{\partial}{\partial u_{\mathfrak{e}}} (MU) \right) \\ &= \sum_{m \geq 1} \text{Tr} \left((MU)^{m-1} \sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} (MU) \right). \end{aligned}$$

Next use the definition of MU to see that

$$\sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} \log \det(I - MU)^{-1} = \sum_{m \geq 1} \text{Tr}((MU)^m). \quad (3.8)$$

Put (3.7) and (3.8) together to get

$$\sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} \log \det(I - MU)^{-1} = \sum_{\mathfrak{e}} u_{\mathfrak{e}} \frac{\partial}{\partial u_{\mathfrak{e}}} \log \zeta_X^{-1}.$$

This implies Theorem 3, up to a constant. The constant must be 0 since both functions are 1 at $u = 0$. ■

COROLLARY 1 (Edge zeta functions of unramified covers). *Suppose Y is a d -fold unramified connected, but possibly irregular and possibly non-normal,*

covering of X . Specialize the d directed edge variables of Y projecting to a single directed edge e of X to u_e and do this for all e of X . With this specialization, the inverse zeta function $\zeta_X^{-1}(u)$ divides $\zeta_Y^{-1}(u)$.

Proof. We know that $\zeta_Y^{-1}(u) = \det(\tilde{I} - \tilde{M}\tilde{U})$, where $\tilde{M}$ denotes the directed edge matrix of Y . Then $\tilde{M}$ is a $(2d|E|) \times (2d|E|)$ matrix. As in the proof of Corollary 1 of Theorem 2 in the last section, we can put $\tilde{M}$ in block form, with blocks of size $(2|E|) \times (2|E|)$. To do this, we need to label a directed edge $\mathbf{f}$ in Y by $(i, \mathbf{e})$, where i denotes the sheet of the starting point of the edge $\mathbf{f}$ and $\mathbf{e}$ denotes the projection by π of $\mathbf{f}$ down to X . So we write $\mathbf{f} = (i, \mathbf{e})$, with $1 \leq i \leq d$ and $\mathbf{e}$ a directed edge of X . Let $\mathbf{g}$ be another directed edge of Y , with $\mathbf{g} = (j, \mathbf{e}')$. Then $\tilde{M}_{\mathbf{f}, \mathbf{g}} = 1$ if $\mathbf{f}$ feeds into $\mathbf{g}$ without backtracking.

CLAIM. $\tilde{M} = (M_{ij})_{1 \leq i, j \leq d}$ such that for each i , $\sum_j M_{ij} = M$.

Proof. To see this, first note that if there is a 1 at $(i, \mathbf{e})$, $(j, \mathbf{e}')$, then there is a 0 at $(i, \mathbf{e})$, $(k, \mathbf{e}')$, for $k \neq j$.

Second, if $M_{\mathbf{e}, \mathbf{e}'} = 1$, then $\mathbf{e}$ feeds into $\mathbf{e}'$. Then on sheet i there is a lift of $\mathbf{e}$ to an edge $\mathbf{f} = (i, \mathbf{e})$ and the edge $(i, \mathbf{e})$ terminates on a unique sheet j . Then $\mathbf{g} = (j, \mathbf{e}')$ is the desired edge of Y so that $\tilde{M}_{\mathbf{f}, \mathbf{g}} = 1$. This completes the proof of the claim. ■

To finish the proof of Corollary 1, proceed exactly as in the proof of Corollary 1 to Theorem 2 in the last section. ■

EXAMPLE 8 (The zeta function of our favorite subgraph X of K_4). Here we check our Theorem 3 and Proposition 2 for the special case of the graph in Fig. 14. We find that

$$\det(I - MU)|_{u_j = u, \text{ all } j = 1, \dots, 10} = Z_X(u)^{-1}.$$

The determinant we need to compute is that of the matrix

$$\begin{pmatrix} -1 & u & 0 & 0 & 0 & 0 & 0 & 0 & u & 0 \\ 0 & -1 & u & 0 & u & 0 & 0 & 0 & 0 & 0 \\ u & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u & 0 & -1 & 0 & u & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & u & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & u & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & u & -1 & 0 & u & 0 \\ 0 & 0 & 0 & 0 & u & 0 & u & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & u \\ 0 & 0 & u & 0 & 0 & 0 & u & 0 & 0 & -1 \end{pmatrix}.$$

The determinant is

$$1 - 4u^3 - 2u^4 + 4u^6 + 4u^7 + u^8 - 4u^{10}.$$

This is equal to

$$\begin{aligned} & (1 - u^2)(1 - u)(1 + u^2)(1 + u + 2u^2)(1 - u^2 - 2u^3) \\ &= (1 - u^2)(1 + u^2 - 4u^3 - u^4 - 4u^5 + 3u^6 + 4u^8). \end{aligned}$$

4. THE PATH ZETA FUNCTION

Let r be the number of generators of the fundamental group Γ of our finite graph X . We will define a zeta function with $(2r)(2r - 1)$ variables. Why so many variables? Our first thought was to create a function of $2r$ variables—two per generator of the fundamental group (one labeled by the generator and one labelled by its inverse). Indeed there is such a function, and it turns out to be a rational function of the $2r$ variables, but outside of graphs such as the one shown in Fig. 16 (set all four variables equal to u^3), the variables cannot be specialized to give the one variable Ihara zeta function, let alone the multivariable edge zeta function of the last section. The difficulty is that in most graphs, there are overlaps in the generators of the fundamental group and it is impossible to account for these overlaps in making specializations of variables attached to just a single generator. It is our desire to specialize to the multivariable edge zeta function that motivates introducing a variable corresponding to each pair of generators of the fundamental group. This gives us enough of a “memory” to be able to account for overlaps in making specializations.

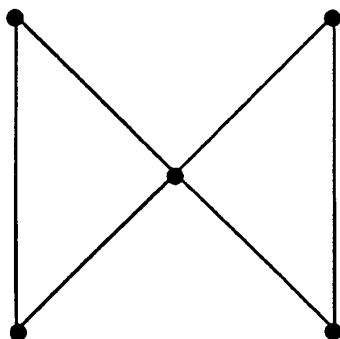


FIG. 16. A graph for which $2r$ variables would suffice.

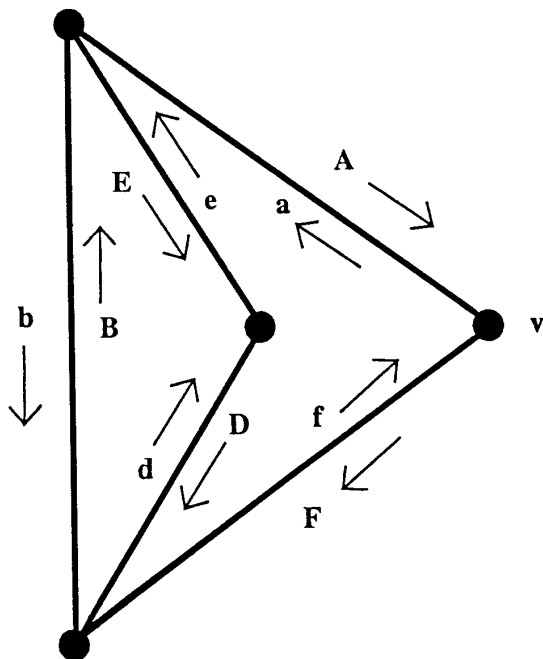


FIG. 17. The favorite example for this section—a relabeling of the graph of Fig. 14.

The path zeta function will be a product over primitive conjugacy classes in the fundamental group. To understand this function, we need to recall the 1-1 correspondence between equivalence classes $[C]$ of backtrack-less, tail-less cycles in X and conjugacy classes in the fundamental group Γ .

EXAMPLE. The favorite example from Section 3 will continue in this section. That is we will again consider the graph of Fig. 14, but we will relabel the edges to eliminate the subscripts as in Fig. 17. We take the base point for the fundamental group to be the vertex v at the start of edges a and F . Two generators for the fundamental group of this graph are

$$P_1 = abf, \quad P_2 = abdeA.$$

A typical element of the fundamental group Γ of this graph is

$$\gamma = P_1 P_2 P_1 P_2^2 P_1^{-1}.$$

A better element of the same conjugacy class is

$$P_2 P_1 P_2^2.$$

There are four possible conjugate versions of this element which do not have some path and its inverse at the outer ends:

$$P_2 P_1 P_2^2, \quad P_1 P_2^3, \quad P_2^3 P_1, \quad P_2^2 P_1 P_2.$$

These are cyclic rearrangements of each other. We shall view these four versions of the element as equivalent. This is the equivalence relation analogous to that between backtrack-less, tail-less cycles in Section 1.

When we eliminate backtracking and tails, the path $P_2 P_1 P_2^2$ gives the reduced cycle $C = bde bfa bde bde$. Similarly $P_1 P_2^3$, $P_2^3 P_1$, $P_2^2 P_1 P_2$ give reduced cycles which are equivalent to C .

DEFINITION. Let $P_1, \dots, P_r$ generate the fundamental group Γ of X . Write $P_{r+1} = P_1^{-1}, \dots, P_{2r} = P_r^{-1}$. We will always order the indices $i = 1, \dots, 2r$ to correspond to this numbering of the generators and their inverses. The *path zeta function* is a function of $(2r)(2r-1)$ sufficiently small complex variables $u_{i,j}$, $1 \leq i, j \leq 2r$, $|i-j| \neq r$,

$$\mathcal{Z}_\Gamma(u) = \prod_{\substack{\{C\}, \text{ primitive} \\ \text{conjugacy} \\ \text{classes in } \Gamma}} (1 - f(C))^{-1}, \quad (4.1)$$

where

$$f(P_{i_1} P_{i_2} \cdots P_{i_n}) = u_{i_1, i_2} u_{i_2, i_3} \cdots u_{i_{n-1}, i_n} u_{i_n, i_1},$$

and we assume that

$$P_{i_1} \neq P_{i_n}^{-1}, \quad P_{ij} \neq P_{ij+1}^{-1} \quad \text{for all } j. \quad (4.2)$$

Note that we can choose an element C of the conjugacy class $\{C\}$ satisfying conditions (4.2) and that $f(C)$ is independent of the choice of C , since different paths

$$C = P_{i_1} P_{i_2} \cdots P_{i_n},$$

satisfying (4.2), are cyclic permutations of each other and the variables u_{ij} commute.

THEOREM 4 (Generalization of Ihara's theorem to the path zeta function). *We take a set of $2r$ indices corresponding to*

$$P_1, \dots, P_r, P_{r+1} = P_1^{-1}, \dots, P_{2r} = P_r^{-1}.$$

Then, for the path zeta function defined by (4.1), we have

$$\mathcal{Z}_\Gamma(u)^{-1} = \det(I - U),$$

where U is a $(2r) \times (2r)$ matrix with ij entry u_{ij} , with the $2r$ additional variables set to 0,

$$u_{i, i+r} = 0 = u_{i+r, i} \quad \text{for } i = 1, \dots, r.$$

Proof. We imitate the proof of Theorem 3. As in (3.4), set

$$L = \sum_{\substack{j, k=1 \\ |j-k| \neq r}}^{2r} u_{j, k} \frac{\partial}{\partial u_{j, k}}. \quad (4.3)$$

Then with $f(C)$ as in (4.1), we have

$$\begin{aligned} L \log \mathcal{Z}_r(u)^{-1} &= \sum_{m \geq 1} \frac{1}{m} \sum_{\substack{C = P_{i_1} P_{i_2} \cdots P_{i_m} \\ \text{primitive} \\ \text{satisfying (4.2)}}} \sum_{j \geq 1} \frac{1}{j} L(f(C)^j) \\ &= \sum_{m \geq 1} \sum_{\substack{C = P_{i_1} P_{i_2} \cdots P_{i_m} \\ \text{satisfying (4.2)}}} f(C), \end{aligned}$$

with the last sum no longer being over primitive C .

As in (3.6), we define

$$N'_m(u) = \sum_{\substack{C = P_{i_1} P_{i_2} \cdots P_{i_m} \\ \text{satisfying (4.2)}}} f(C). \quad (4.4)$$

So it follows that

$$L \log \mathcal{Z}_r(u)^{-1} = \sum_{m \geq 1} N'_m(u). \quad (4.5)$$

Now, as in Lemma 5, we have

$$N'_m(u) = \text{Tr}(U^m), \quad (4.6)$$

where U is the matrix defined in the statement of Theorem 4. To see (4.6), look, for example, at

$$(U^3)_{i, i} = \sum_{j, k=1}^{2r} u_{i, j} u_{j, k} u_{k, i} = \sum_{\substack{C = P_i P_j P_k \\ P_i \neq P_j^{-1}, P_j \neq P_k^{-1}, P_k \neq P_i^{-1}}} f(C).$$

So combining (4.5) and (4.6) yields

$$L \log \mathcal{Z}_r(u)^{-1} = \sum_{m \geq 1} \text{Tr}(U^m). \quad (4.7)$$

Now we turn to the other side of the equation in Theorem 4. Again we can use Lemma 6 to see that

$$\begin{aligned} L \log \det(I - U)^{-1} &= L \operatorname{Tr} \left(\sum_{m \geq 1} \frac{1}{m} U^m \right) \\ &= \sum_{m \geq 1} \operatorname{Tr}(U^{m-1} L U) \\ &= \sum_{m \geq 1} \operatorname{Tr}(U^m), \end{aligned}$$

since

$$(LU)_{s,t} = \sum_{\substack{i,j=1 \\ |i-j| \neq r}}^{2r} u_{i,j} \frac{\partial}{\partial u_{i,j}} U_{s,t} = u_{s,t}.$$

Combining this with (4.7), gives

$$L \log \det(I - U)^{-1} = L \log \mathcal{Z}_r(u)^{-1}.$$

This implies the theorem up to a constant and the constant is 0, since both functions are 1 at $u = 0$. ■

Note. If we set each non-zero $u_{ij} = w_i$ (for all $j \neq i \pm r$), then we get the $2r$ variable zeta function mentioned in the first paragraph of this section.

It is harder to obtain the specialization theorem which says how we get the edge zeta function by specializing the variables in the path zeta function. First we need to normalize the generators of Γ to be able to specialize the variables correctly.

Normalization of the Generators of the Fundamental Group. The aim is to ensure that no generator (or its inverse) overlaps more than half of another at the beginning or end. We will also assume that the generators have no backtracking. The normalization method is as follows. Suppose $P_1, \dots, P_r$ generate the fundamental group Γ of X . If $P_2 = C_1 C_2$ and $P_1 = C_1 C_3$, where the length of C_1 is more than half the length of P_1 ; i.e.,

$$v(C_1) > \frac{1}{2}v(P_1).$$

Then replace P_2 by $R_2 = P_1^{-1} P_2 = C_3^{-1} C_1^{-1} C_1 C_2 = C_3^{-1} C_2$. This new path $R_2 = C_3^{-1} C_2$ has length less than that of P_2 , since

$$v(C_1) > v(C_3).$$

Do the same construction to eliminate overlaps on the other side. Perform the same construction for all pairs of generators P_i, P_j and pairs

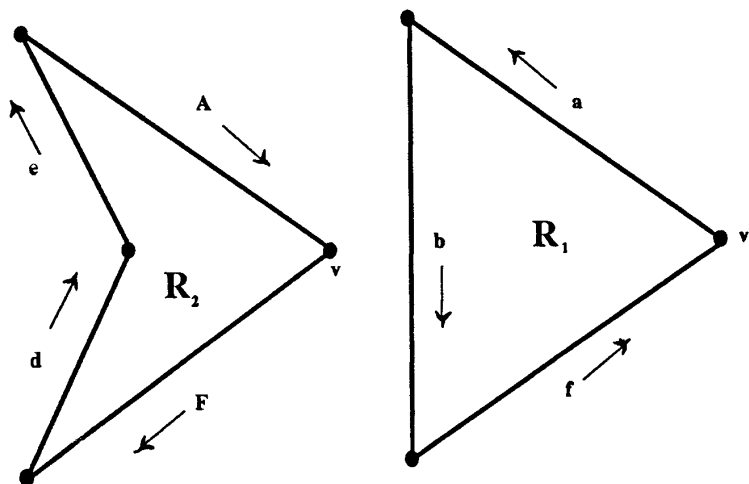


FIG. 18. Two normalized generators of the fundamental group of the graph in Fig. 15.

of generators and inverses P_i, P_j^{-1} . Continue the process on the new generators. Since you keep decreasing the lengths, the process must terminate.

EXAMPLE. Consider the example in Fig. 17 again with

$$P_1 = abf, \quad P_2 = abdeA$$

generating the fundamental group. Since P_1 and P_2 overlap in more than half of P_1 , we need to change the generators to

$$R_1 = P_1, \quad R_2 = P_1^{-1}P_2 = F(BAab)deA = FdeA.$$

These generators are pictured in Fig. 18.

THEOREM 5 (The path zeta function specializes to the edge zeta function). *Assume that the generators of the fundamental group are normalized as just discussed. Specialize the variables u_{ij} , $i \neq j+r$, $j \neq i+r$, as follows. Suppose that*

$$P_i = e_1 \cdots e_l f_m^{-1} f_{m-1}^{-1} \cdots f_1^{-1}, \quad P_j = f_1 f_2 \cdots f_m g_1 g_2 \cdots g_n,$$

where $e_l \neq g_1^{-1}$. Specialize u_{ij} to

$$u_{ij}|_{\text{specialized}} = u_{e_1} \cdots u_{e_l} (u_{f_1} \cdots u_{f_m})^{-1}.$$

Then

$$\mathcal{Z}_F(u)|_{\text{specialized}} = \zeta_X(u).$$

The variables u on the right are those associated to directed paths. The renormalization guarantees that $m \leq l$ and $m \leq n$.

Proof. The specialization has been chosen to eliminate all backtracking and tails. Suppose C is a closed path with no backtracking or tails. Then C corresponds to an element $P_{i_1}P_{i_2}\cdots P_{i_n}$ of the fundamental group with $P_{i_1} \neq P_{i_n}^{-1}$ and $P_{i_j} \neq P_{i_{j+1}}^{-1}$ for all j . The element $P_{i_1}P_{i_2}\cdots P_{i_n}$ of the fundamental group is not unique, but its conjugacy class is. We then have

$$\begin{aligned} f(C)|_{\text{specialized}} &= u_{i_1, i_2} u_{i_2, i_3} \cdots u_{i_{n-1}, i_n} u_{i_n, i_1} |_{\text{specialized}} \\ &= \prod_{e \text{ in } C} u_e. \end{aligned}$$

This is exactly what is needed to prove Theorem 5. To see this statement, look at the following example. ■

EXAMPLE FOR THE GRAPH IN FIG. 17. Choose the normalized generators R_1 and R_2 as described in Fig. 18. Let us work out the various specializations in detail for this example. They change the matrix U to

$$U|_{\text{specialized}} = \begin{pmatrix} u_a u_b u_f & u_a u_b u_F^{-1} & 0 & u_a u_b u_f \\ u_F u_d u_e u_a^{-1} & u_F u_d u_e u_A & u_F u_d u_e u_A & 0 \\ 0 & u_F u_B u_A & u_F u_B u_A & u_F u_B u_a^{-1} \\ u_a u_E u_D u_f & 0 & u_a u_E u_D u_F^{-1} & u_a u_E u_D u_f \end{pmatrix}.$$

To see how we obtained u_{12} , for example, note that we must look at the path $R_1 = abf$ and $R_2 = FdeA$. Now there is an overlap in $(abf)(FdeA)$. So we must specialize u_{12} by writing not $u_a u_b u_f$ but $u_a u_b u_F^{-1}$.

Using these specializations, we can compute the specialization of $f(C)$ for any path C . For example, the path $R_2 R_1$ yields

$$f(R_2 R_1)|_{\text{specialized}} = (u_{21} u_{12})|_{\text{specialized}} = (u_F u_d u_e u_a^{-1})(u_a u_b u_F^{-1}) = u_d u_e u_b.$$

The edge variable product $u_d u_e u_b$ corresponds to the path deb which is what remains of $R_2 R_1 = FdeAabf$ when backtracking and tails are eliminated.

If we specialize all the directed edge variables to be equal to u , we obtain the matrix

$$U' = \begin{pmatrix} u^3 & u & 0 & u^3 \\ u^2 & u^4 & u^4 & 0 \\ 0 & u^3 & u^3 & u \\ u^4 & 0 & u^2 & u^4 \end{pmatrix}.$$

Then we compute

$$\det(I - U') = (1 + u)(1 - u)^2 (1 + u^2)(1 + u + 2u^2)(1 - u^2 - 2u^3).$$

This is the Ihara zeta function of the graph in Fig. 17, just as Theorem 5 said it would be.

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